

Will Maberry

 will-maberry |  willmaberry.com

EDUCATION

Mississippi State University

M.S. Vet. & Biomed. Science Aug. 2026 – May 2028
Computational Biology Concentration
Fully funded research appointment via USDA ARS
Research Advisor: Dr. Isaac Jumper

The University of Texas at Arlington (UTA)

B.S. Computer Science Aug. 2022 – May 2026
Summa Cum Laude; GPA: 3.8+
Fundamentals of AI Certificate

RESEARCH INTERESTS

Spatiotemporal forecasting; computational epidemiology;
geospatial machine learning; biosurveillance systems

TECHNICAL SKILLS

Forecasting & Modeling

Spatiotemporal forecasting; leakage-safe ML pipelines;
rare-event modeling; backtesting; ablation studies;
hyperparameter optimization

Machine Learning

PyTorch; TensorFlow; Scikit-learn; Imbalanced-learn;
Optuna; AutoGluon

Geospatial & Scientific Computing

GeoPandas; Google Earth Engine; raster processing

Programming & Data Engineering

Python; C; Java; SQL; Bash; Pandas; NumPy

Pipelines & Deployment

FastAPI; Flask; Docker; automated forecasting pipelines;
artifact persistence

Visualization

Matplotlib; Folium/Leaflet; interactive dashboards

SELECTED PROJECTS

American Sign Language (ASL) Recognizer

- Developed a real-time ASL recognition pipeline for gesture classification and inference.
- Built a custom dataset of **2000+ samples** for multi-class hand-sign recognition.
- Integrated interactive visualization and text-to-speech for real-time sign interpretation, achieving **95% accuracy**.

Algorithm Learning Platform

- Developed an interactive educational web platform visualizing **20+ algorithms and data structures** (sorting, graph traversal, heaps, and trees) for classroom instruction and self-paced learning.
- Adopted by UTA faculty as supplemental instructional material, supporting **50+ students per semester**.

PUBLICATIONS

Blackstock, J., Maberry, W., et al.

“A forward-looking high pathogenicity avian influenza monthly risk model for counties in the conterminous United States”
Manuscript submitted to Scientific Reports (May 2026)

EXPERIENCE

USDA ARS — Research Assistant Jul. 2025 – Present

- Developing a **forward-looking, national-scale** county-month HPAI forecasting framework for the continental U.S. using environmental, agricultural, and geospatial data streams.
- Implementing **spatiotemporal machine learning models** over county adjacency networks to capture cross-county disease propagation and regional outbreak dynamics.
- Engineering **leakage-safe forecasting pipelines** with forward-only data truncation, forecast-month feature generation, persisted forecast artifacts, and automated monthly backtesting.
- Conducting **model architecture sweeps, ablation studies, and post-model optimization**.
- Designing a **decision-support web application** with county-level risk visualizations, translating forecast outputs into interpretable biosurveillance insights.

USDA ARS — Research Intern May 2025 – Jul. 2025

- Developed national-scale county-level HPAI risk classification and pseudo-forecasting models using environmental and climate covariates across the continental U.S.
- Achieved **75%+ balanced accuracy** on highly imbalanced (~1% positive) outbreak data spanning 2022–2024.
- Presented research findings to the **USDA ARS Chief Scientist** and multiple National Program Leaders (NPLs) to support disease biosurveillance.
- Received the **Judge’s Choice Award** at UTA College of Engineering Innovation Day 2026.

OpenAI — Engagement Manager Jun. 2024 – Jun. 2026

- Scaled OpenAI’s official Discord community to **850k+ users** (680% growth) through technical programming, online events, and community engagement initiatives.

TEACHING & COMMUNITY

- Teaching Assistant (Operating Systems)**: Supported 120+ students through office hours and review sessions.
- OpenAI Community Volunteer**: Supported global AI education initiatives and technical discussions (2022–2024).
- Association for Computing Machinery (ACM)**: Education Director for 1500+ member organization; led 15+ professional and educational workshops.
- The Wesley @ UTA**: Sole student representative on Board of Directors overseeing \$100k/year organizational budget.